Capstone limitations and revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete evaluation and revision record.

Keep frozen predictions, condition errors, original scores and proposed changes together.

Keep one habit: Recall → W1–W3/W5 → independent evaluation → project revision history → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone controlled experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Freeze the Week 34 candidate: max(0, a*x), initial a = 1, rate 0.2, two full-batch updates on (1,2),(2,4), no bias. The final weight is 2. The baseline is unchanged at a = 1.

  Now open final rows F1=(5,10), F2=(6,12), F3=(-1,-2), F4=(-2,-4). These fictional rows deliberately include a condition absent from the positive **training** and validation **input**s.

  Opening them is an evaluation event, not permission to tune on them while retaining an independent-test claim.

- Write raw **prediction**s for both procedures before summarising. A low overall error can hide a systematic problem in a condition. These four constructed examples do not estimate prevalence or real-world performance.

Capstone controlled experiments

> **My remembered idea, example and correction**

Documenting a network experiment

> **My remembered idea, example and correction**

Uncertainty and failure analysis

> **My remembered idea, example and correction**

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Frozen main candidate ReLU(2x), baseline ReLU(x). Four constructed evaluation rows F1(5, 10), F2(6, 12), F3(−1, −2), F4(−2, −4). Baseline predictions 5, 6, 0, 0; residuals−5, −6, 2, 4; squares 25, 36, 4, 16; MSE 81/4=20.25. Candidate predictions 10, 12, 0, 0; residuals 0, 0, 2, 4; squares 0, 0, 4, 16; MSE 20/4=5.

  Strict overall improvement passes; candidate positive-input MSE 0 on two cases, negative-input MSE 10 on two. ReLU output cannot be negative for any weight, so more epochs do not remove that representational limit. Separate supported x−3, target−6: ReLU(2x)0, residual 6, square 36; signed 2x gives−6, error 0 on a revealed case.

  Signed output is a proposed architecture revision needing appropriate fitting/selection and new reserved final evidence. Preserve original MSE 5. Separate picture centred 7/8=87.5%, shifted 3/8=37.5%; training on exposed shifted evaluation cases turns their later scores into development checks, not independent robustness evidence. Published cases already seen during choices are teaching/exposed cases.

- For F1–F4 calculate candidate **residual**s and squared errors.

> **My reasoning and evidence**
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MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Frozen main candidate ReLU(2x), baseline ReLU(x). Four constructed evaluation rows F1(5, 10), F2(6, 12), F3(−1, −2), F4(−2, −4). Baseline predictions 5, 6, 0, 0; residuals−5, −6, 2, 4; squares 25, 36, 4, 16; MSE 81/4=20.25. Candidate predictions 10, 12, 0, 0; residuals 0, 0, 2, 4; squares 0, 0, 4, 16; MSE 20/4=5.

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- Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Frozen main candidate ReLU(2x), baseline ReLU(x). Four constructed evaluation rows F1(5, 10), F2(6, 12), F3(−1, −2), F4(−2, −4). Baseline predictions 5, 6, 0, 0; residuals−5, −6, 2, 4; squares 25, 36, 4, 16; MSE 81/4=20.25. Candidate predictions 10, 12, 0, 0; residuals 0, 0, 2, 4; squares 0, 0, 4, 16; MSE 20/4=5.

  Strict overall improvement passes; candidate positive-input MSE 0 on two cases, negative-input MSE 10 on two. ReLU output cannot be negative for any weight, so more epochs do not remove that representational limit. Separate supported x−3, target−6: ReLU(2x)0, residual 6, square 36; signed 2x gives−6, error 0 on a revealed case.

  Signed output is a proposed architecture revision needing appropriate fitting/selection and new reserved final evidence. Preserve original MSE 5. Separate picture centred 7/8=87.5%, shifted 3/8=37.5%; training on exposed shifted evaluation cases turns their later scores into development checks, not independent robustness evidence. Published cases already seen during choices are teaching/exposed cases.

- Why cannot extra **training** alone make this ReLU **output** negative? Name a proposed change and the **evidence** it would require.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Frozen main candidate ReLU(2x), baseline ReLU(x). Four constructed evaluation rows F1(5, 10), F2(6, 12), F3(−1, −2), F4(−2, −4). Baseline predictions 5, 6, 0, 0; residuals−5, −6, 2, 4; squares 25, 36, 4, 16; MSE 81/4=20.25. Candidate predictions 10, 12, 0, 0; residuals 0, 0, 2, 4; squares 0, 0, 4, 16; MSE 20/4=5.

  Strict overall improvement passes; candidate positive-input MSE 0 on two cases, negative-input MSE 10 on two. ReLU output cannot be negative for any weight, so more epochs do not remove that representational limit. Separate supported x−3, target−6: ReLU(2x)0, residual 6, square 36; signed 2x gives−6, error 0 on a revealed case.

  Signed output is a proposed architecture revision needing appropriate fitting/selection and new reserved final evidence. Preserve original MSE 5. Separate picture centred 7/8=87.5%, shifted 3/8=37.5%; training on exposed shifted evaluation cases turns their later scores into development checks, not independent robustness evidence. Published cases already seen during choices are teaching/exposed cases.

- After inspecting F1–F4, a student changes to **prediction** 2*x and reports “new final MSE 0”. Explain the accurate label and the next evaluation step.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Evaluate new evidence independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6).

  Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Compare with the baseline for Capstone limitations and **revision**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Capstone limitations and **revision**. Proposed explanation: "Passing a criterion means no failures remain". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12), (-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.
- Centred slice: 7/8 correct. Shifted slice: 3/8 correct. The network and label rubric are frozen.
- Calculate the slice rates and explain the claim **revision**. If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record.

  State exactly what this example supports before linking it to the wider capstone claim.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/35

Capstone limitations and revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Centred is 87.5%; shifted is 37.5%.



2. Revise the claim

Do not describe the network as shift-robust on this evidence.



3. Plan a fair revision

Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

CHANGE ONE THING

Train the revision on all eight failed-slice evaluation patterns and rescore them.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.
- Which labelled step matches this detail? Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.
- 1. Compute slice rates
- 2. Revise the claim
- 3. Plan a fair **revision**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Plan a fair **revision** Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Changed-case question and optional hint

- Changed-case practice: Train the **revision** on all eight failed-slice evaluation patterns and rescore them. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Treat the score as exposed-case development, not independent **evidence** of shift robustness.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Treat the score as exposed-case development, not independent **evidence** of shift robustness. The evaluation records now influenced learning and model selection.

Repair and retain the record

Review Capstone controlled experiments (ai-5/34). Goal: Trace matched two-update learning-rate experiments. Short practice: Compute the two correct counts and identify the changed case.

Why would changing B's **training** data weaken the conclusion about its added connection?. Return to this week and explain what you changed.

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Capstone limitations and revision: the case

AI ACADEMY / AI-5/35 · 01 · LOOK AT THE FACTS

Capstone limitations and revision: the case

The situation

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Supplied evidence

Centred slice: 7/8 correct. Shifted slice: 3/8 correct. The network and label rubric are frozen.

Explain capstone limitations and revision with a traceable toy-network case and a bounded conclusion.

Follow the labelled picture with the matching explanation. Open the vector diagram

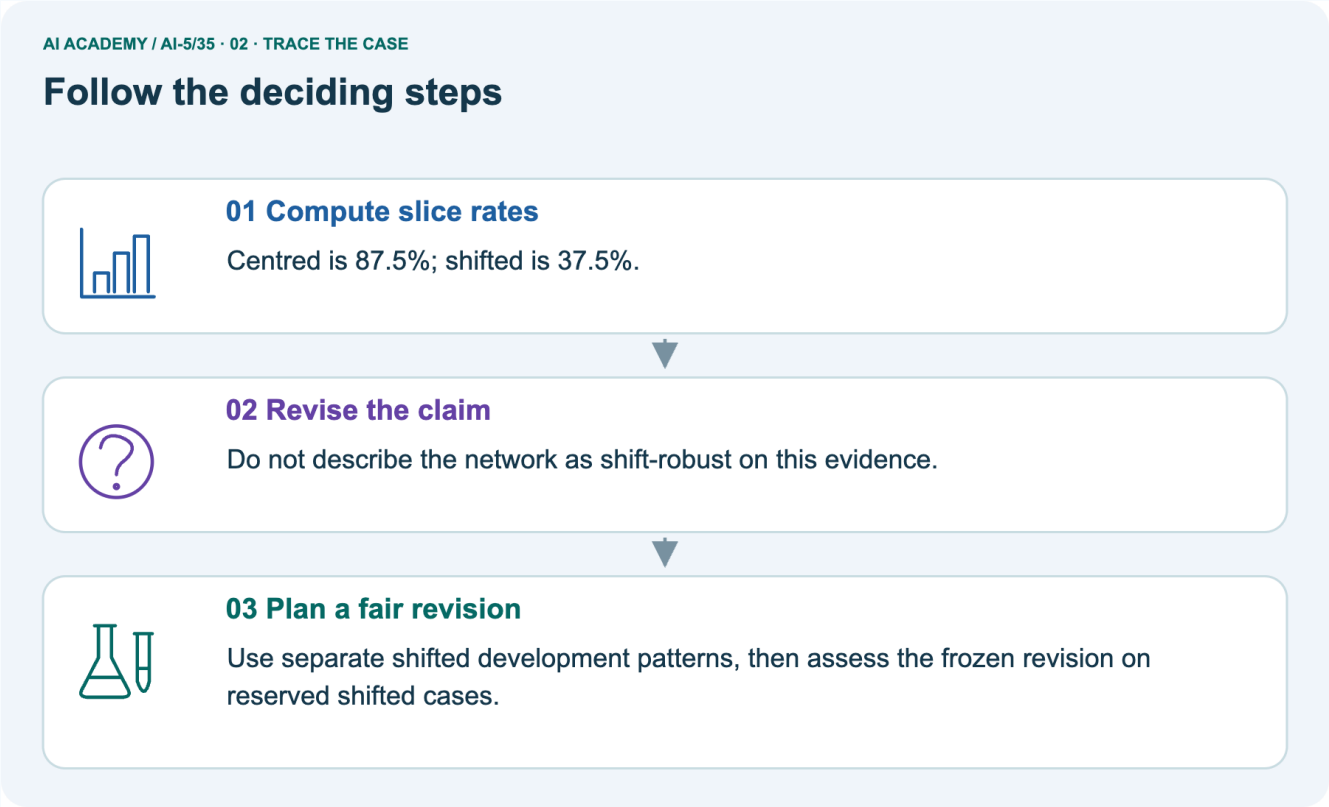
Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim. Supplied facts: Centred slice: 7/8 correct.

Shifted slice: 3/8 correct. The network and label rubric are frozen.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



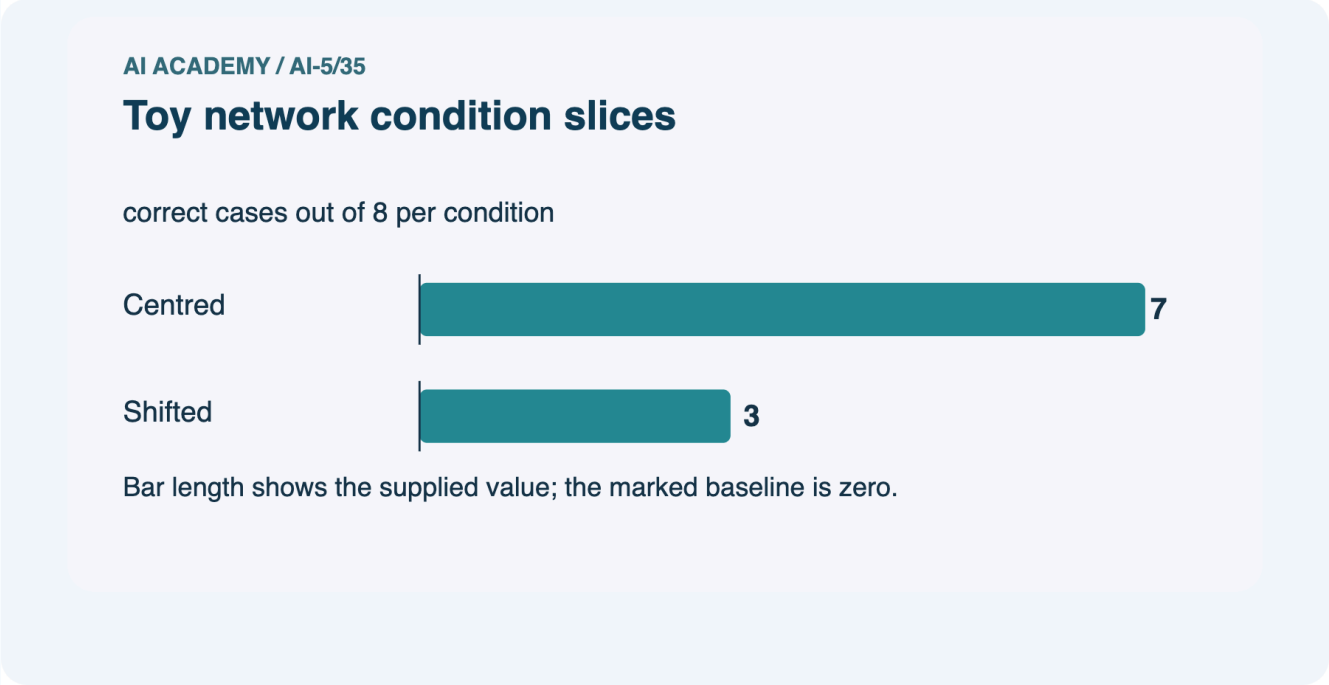
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Centred is 87.5%; shifted is 37.5%. → Revise the claim: Do not describe the network as shift-robust on this **evidence**. → Plan a fair **revision**: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Toy network condition slices



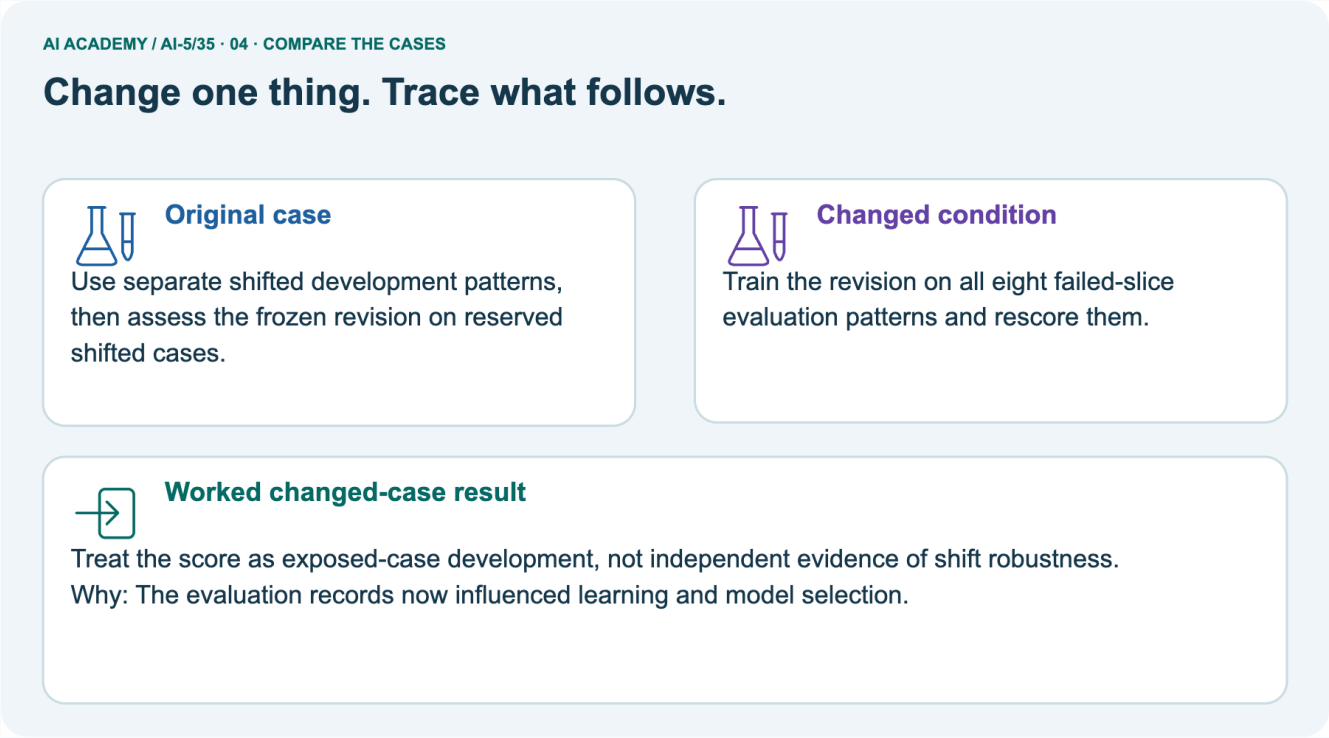
Follow the labelled picture with the matching explanation. Open the vector diagram

Toy network condition slices. Units: correct cases out of 8 per condition. Centred: 7. Shifted: 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

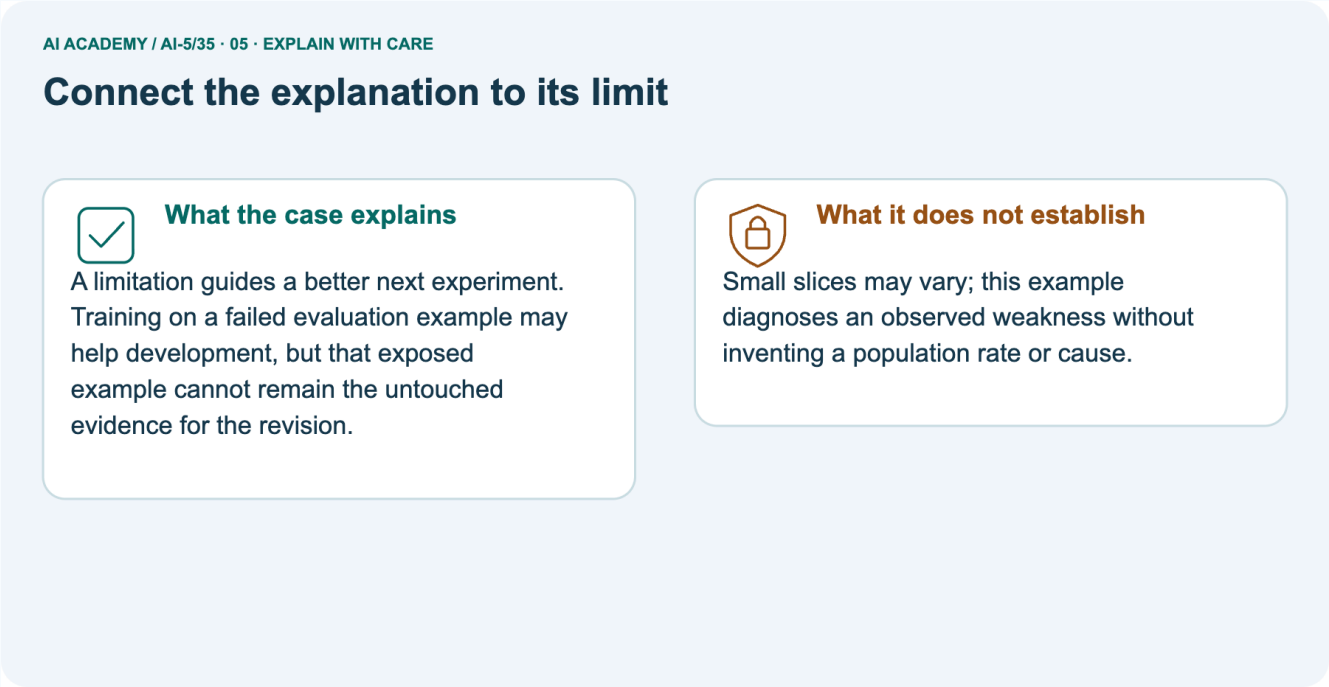
Original case: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases. Changed condition: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

Expected result: Treat the score as exposed-case development, not independent **evidence** of shift robustness. Reason: The evaluation records now influenced learning and model selection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

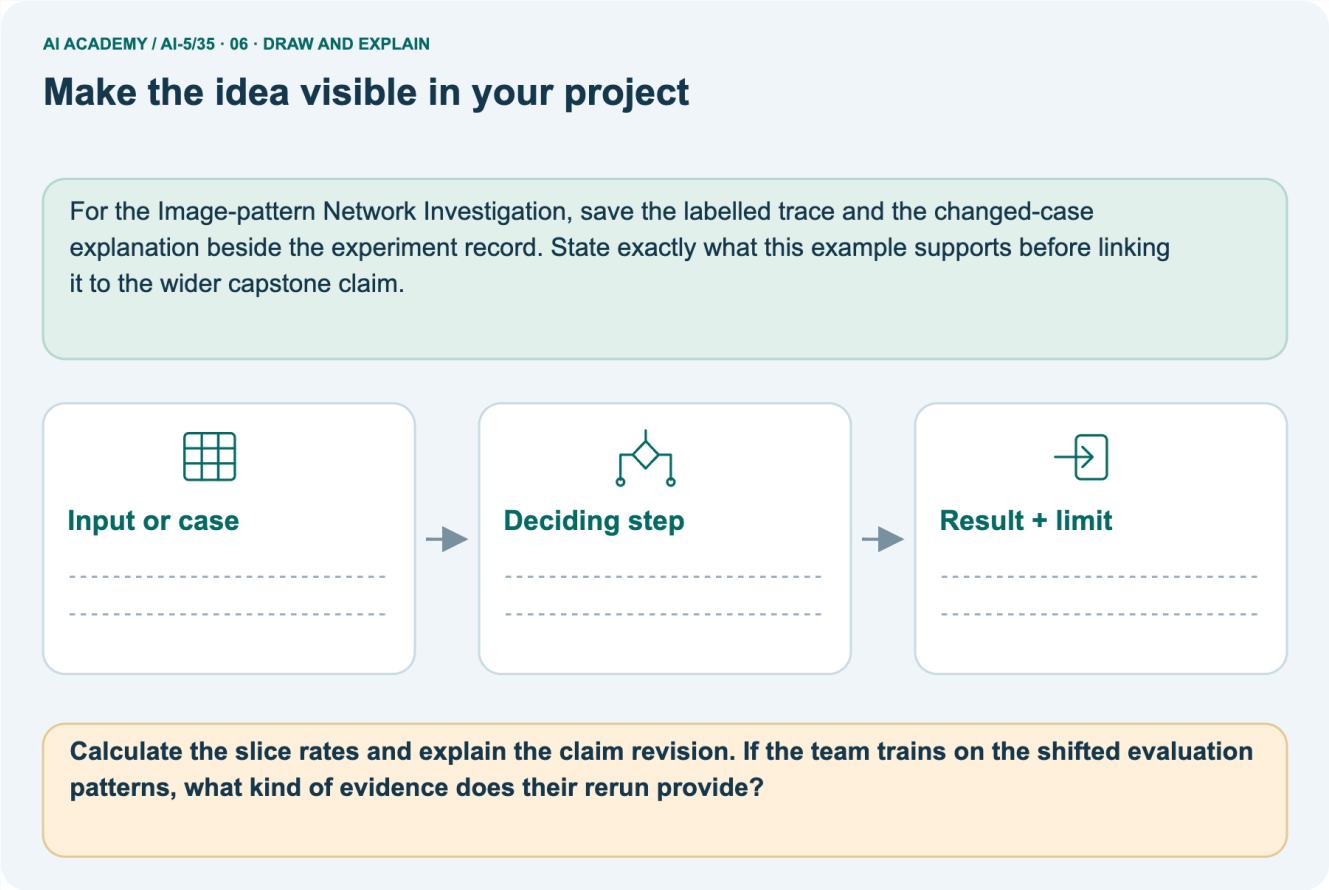
A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Limit: Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the slice rates and explain the claim **revision**.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone limitations and revision: the case

02. Follow the deciding steps

03. Toy network condition slices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.